

TUTORIAL – 14

Selenium Web Automation

Aim:

To create Selenium automation scripts in Python for browser navigation, form handling, web element interaction and validation testing.

Description:

Selenium WebDriver allows testers to automate web browsers using Python.

In this practical, students will use **PyCharm + Python + Selenium + Chrome** to automate login, interact with web elements, select a product, navigate to the cart and validate the application.

Problem Statement:

Create a Selenium automation script that:

1. Opens a website.
2. Handles the login form.
3. Enters username and password.
4. Clicks the login button.
5. Validates the product page.
6. Selects a product.
7. Opens the cart.
8. Validates the URL and page title.

Implementation Details

Parameter	Details
Project Name	Selenium Web Automation
Application/Software	SauceDemo
Module Name	Login and Product Navigation
SDLC Model	Agile
Programming Language	Python
IDE/Tool Used	PyCharm
Automation Tool	Selenium WebDriver
Browser	Google Chrome
Input/Test Data	Username, Password, Product

Automation Environment Configuration

Parameter	Details
Python Version	Python 3.x
Selenium Version	Installed using pip
Browser	Google Chrome
WebDriver	Selenium-managed Chrome WebDriver
IDE Used	PyCharm
Target Website URL	https://www.saucedemo.com/

Automation Scenario Table

Scenario ID	Scenario	Module/Page	Expected Result
1	Browser Navigation	Login Page	Website opens
2	Form Handling	Login	Credentials entered
3	Web Element Interaction	Products	Product selected
4	Validation Testing	Products/Cart	URL/title validated

Procedure

Step 1 – Create PyCharm Project

Create:

SoftwareTesting

Step 2 – Create Python File

tutorial14.py

Step 3 – Install Selenium

pip install selenium

Step 4 – Write Program

Import:

from selenium import webdriver

from selenium.webdriver.common.by import By

Step 5 – Start Chrome

```
driver = webdriver.Chrome()  
driver.maximize_window()
```

Step 6 – Open Website

```
driver.get("https://www.saucedemo.com/")
```

Step 7 – Enter Login Data

```
driver.find_element(  
    By.ID, "user-name"  
)send_keys("standard_user")
```

```
driver.find_element(  
    By.ID, "password"  
)send_keys("secret_sauce")
```

Step 8 – Click Login

```
driver.find_element(  
    By.ID, "login-button"  
)click()
```

Step 9 – Validate

```
assert "inventory" in driver.current_url
```

Step 10 – Add Product

```
driver.find_element(  
    By.ID,  
    "add-to-cart-sauce-labs-backpack"  
)click()
```

Step 11 – Open Cart

```
driver.find_element(  
    By.CLASS_NAME,  
    "shopping_cart_link"  
)click()
```

Step 12 – Close Browser

```
driver.quit()
```

Source Code

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
```

```
driver = webdriver.Chrome()
driver.maximize_window()
```

```
wait = WebDriverWait(driver, 10)
```

```
try:
```

```
    # Browser Navigation
    driver.get("https://www.saucedemo.com/")
```

```
    print("Page Title:", driver.title)
```

```
    # Form Handling
    wait.until(
        EC.visibility_of_element_located(
            (By.ID, "user-name")
        )
    ).send_keys("standard_user")
```

```
    driver.find_element(
        By.ID, "password"
    ).send_keys("secret_sauce")
```

```
    # Web Element Interaction
    driver.find_element(
        By.ID, "login-button"
    ).click()
```

```
    # Validation
    wait.until(
        EC.url_contains("inventory")
    )
```

```

assert "inventory" in driver.current_url

print("Login Validation: PASS")

# Product Interaction
wait.until(
    EC.element_to_be_clickable(
        (By.ID, "add-to-cart-sauce-labs-backpack")
    )
).click()

print("Product Selection: PASS")

# Cart Navigation
driver.find_element(
    By.CLASS_NAME,
    "shopping_cart_link"
).click()

assert "cart" in driver.current_url

print("Cart Navigation: PASS")
print("All Selenium Web Automation Tests Passed")

finally:
    driver.quit()

```

Automation Test Case Details

Field	Details
Test Case ID	TC-14-01
Script Name	tutorial14.py
Module Name	Login/Product
Test Scenario	Automate login and product selection
Test Data	standard_user / secret_sauce

Expected Result	Login and product selection successful
Actual Result	Successfully executed
Status	Pass

Automation Script Execution

Step No.	Automation Action	Expected Result	Actual Result	Status
1	Open website	Login page opens	Opened	Pass
2	Enter username	Username entered	Entered	Pass
3	Enter password	Password entered	Entered	Pass
4	Click login	Products page opens	Opened	Pass
5	Add product	Product added	Added	Pass
6	Open cart	Cart opens	Opened	Pass

Web Element Identification Table

Element Name	Locator Type	Locator Value	Purpose
Username	ID	user-name	Enter username
Password	ID	password	Enter password
Login Button	ID	login-button	Submit login
Product	ID	add-to-cart-sauce-labs-backpack	Add product
Cart	Class Name	shopping_cart_link	Open cart

Validation Results

Validation	Expected Result	Actual Result	Status
Page Title	SauceDemo title displayed	Displayed	Pass
URL	inventory URL	inventory URL	Pass
Form Submission	Login successful	Successful	Pass

Success/Error Message	No login error	No error	Pass
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Automation Execution Summary

Metric	Value
Total Test Scripts	1
Executed	1
Passed	1
Failed	0
Execution Time	_____

Observation

Sr. No.	Observation	Remarks
1	Website opened through Selenium	Successful
2	Login form handled	Successful
3	Product element identified	Successful
4	Product added to cart	Successful
5	URL validation performed	Successful

Output

Students should capture:

1. PyCharm source code
2. Login page
3. Product page
4. Added product
5. Cart page
6. Console output

Questionnaire

Q1. What is Selenium?

Answer: Selenium is an open-source framework used for automating web browsers and testing web applications.

Q2. What is Selenium WebDriver?

Answer: Selenium WebDriver is an API used to control and interact with web browsers programmatically.

Q3. What is ChromeDriver?

Answer: ChromeDriver is the browser driver component that enables Selenium to communicate with Google Chrome.

Q4. Which method is used to enter text into a textbox?

Answer: `send_keys()`

Q5. Which method is used to click a button?

Answer: `click()`